



The Iby and Aladar Fleischman
Faculty of Engineering
Tel Aviv University

Drone Identification Using Micro-Doppler

Project Number: 24-1-1-3174

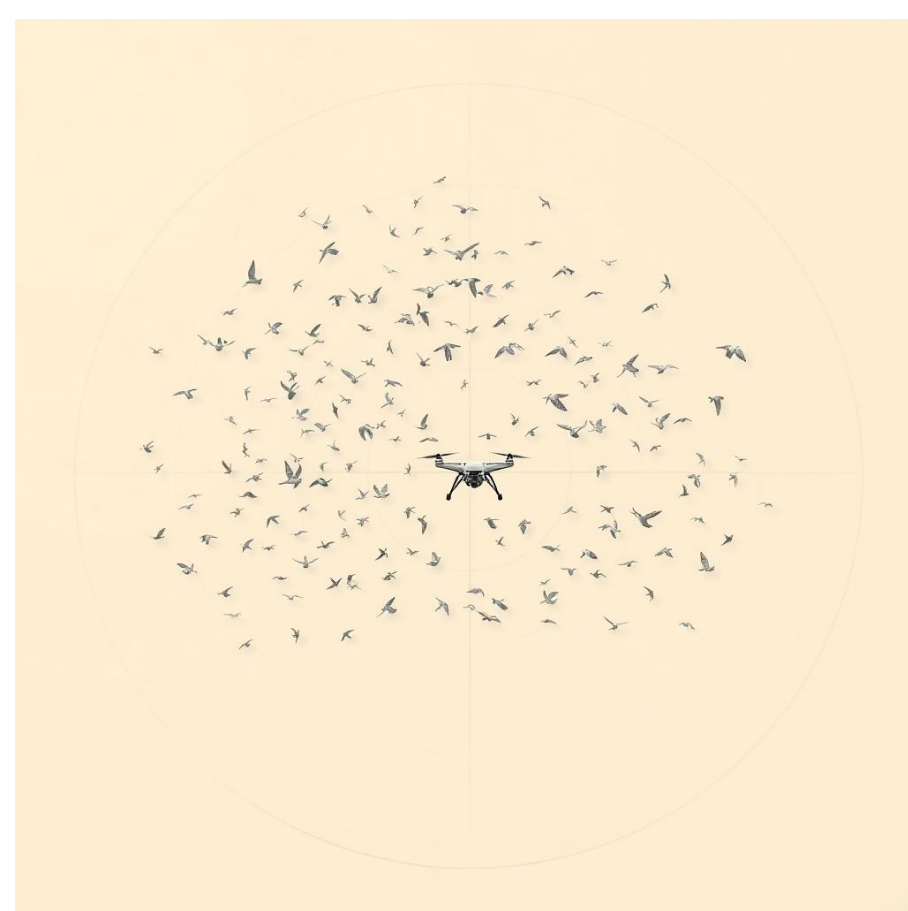
Names: Amit Stein, Niv Avivi

Supervisor : Omer Tzidki

Motivation

Growing drone use demands detection Beyond optical/thermal limits- in bad weather, urban clutter, or among other UAVs. A TAU Radar Lab study showed that arranging dipole tags on rotor blades yields unique 8-bit micro-Doppler codes.

This method leverages enhanced Scattering to overcome low RCS and reliably identify individual drones under all conditions.

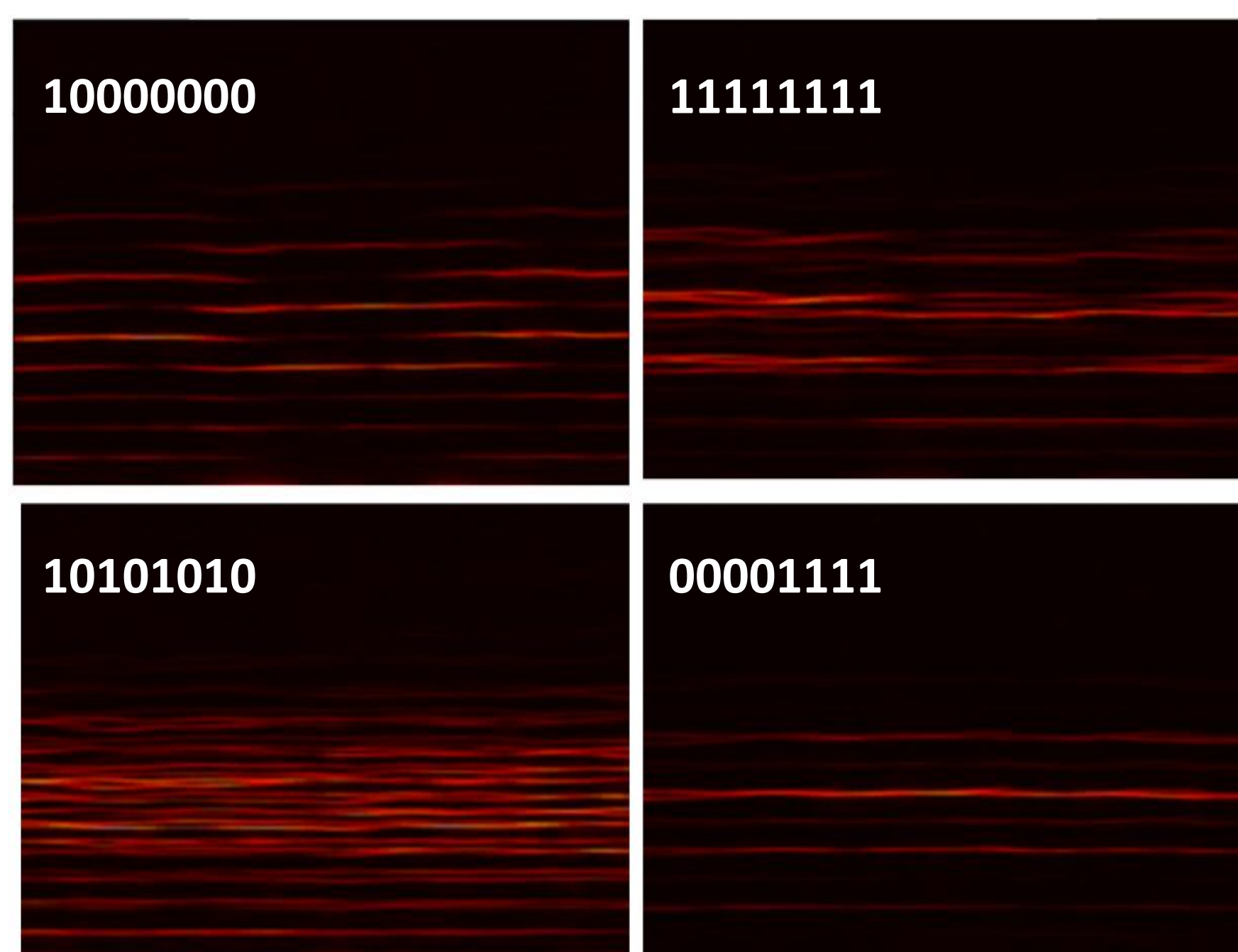


Introduction

In our project, we implement the Classifier component of this research, using measured micro-Doppler signatures. Our objective is to accurately classify noisy micro-Doppler signals at low SNRs ratios. To achieve this, we employ a convolutional neural network (CNN) trained with a synthetic data pipeline and extensive augmentation, demonstrating the robustness of our approach in noisy SNR environments.



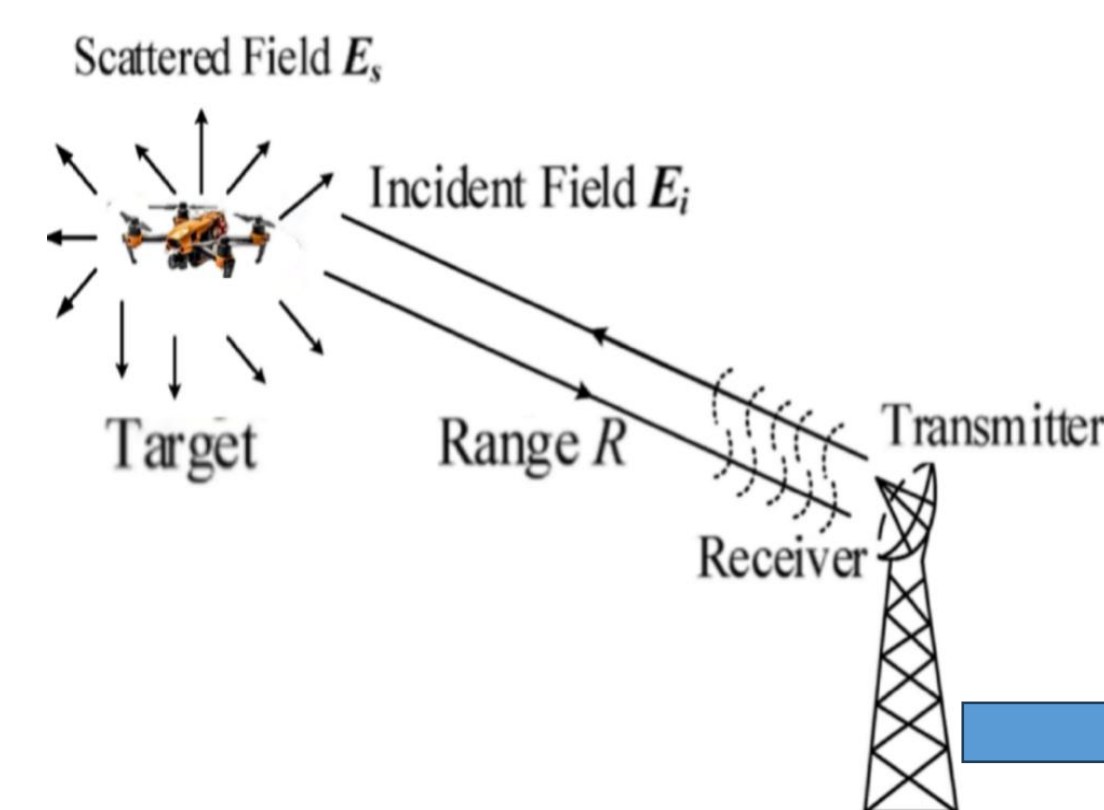
Spectrograms for different microdopler signatures



Example of a drone from class 11001010



The dipole tag



Implementation

We were given six 10-second signals (50 001 complex samples) for each of 34 tag configurations indoors.

Segmentation & shifts:

Split each signal into 7,700-sample segments with 50 % overlap and From each segment, generate seven time-shifted versions (plus the original).

Spectrogram & split:

Compute a normalized STFT (256-sample Hann window, 50 % overlap) and stratify into 70 % train, 20 % validation, 10 % hold-out.

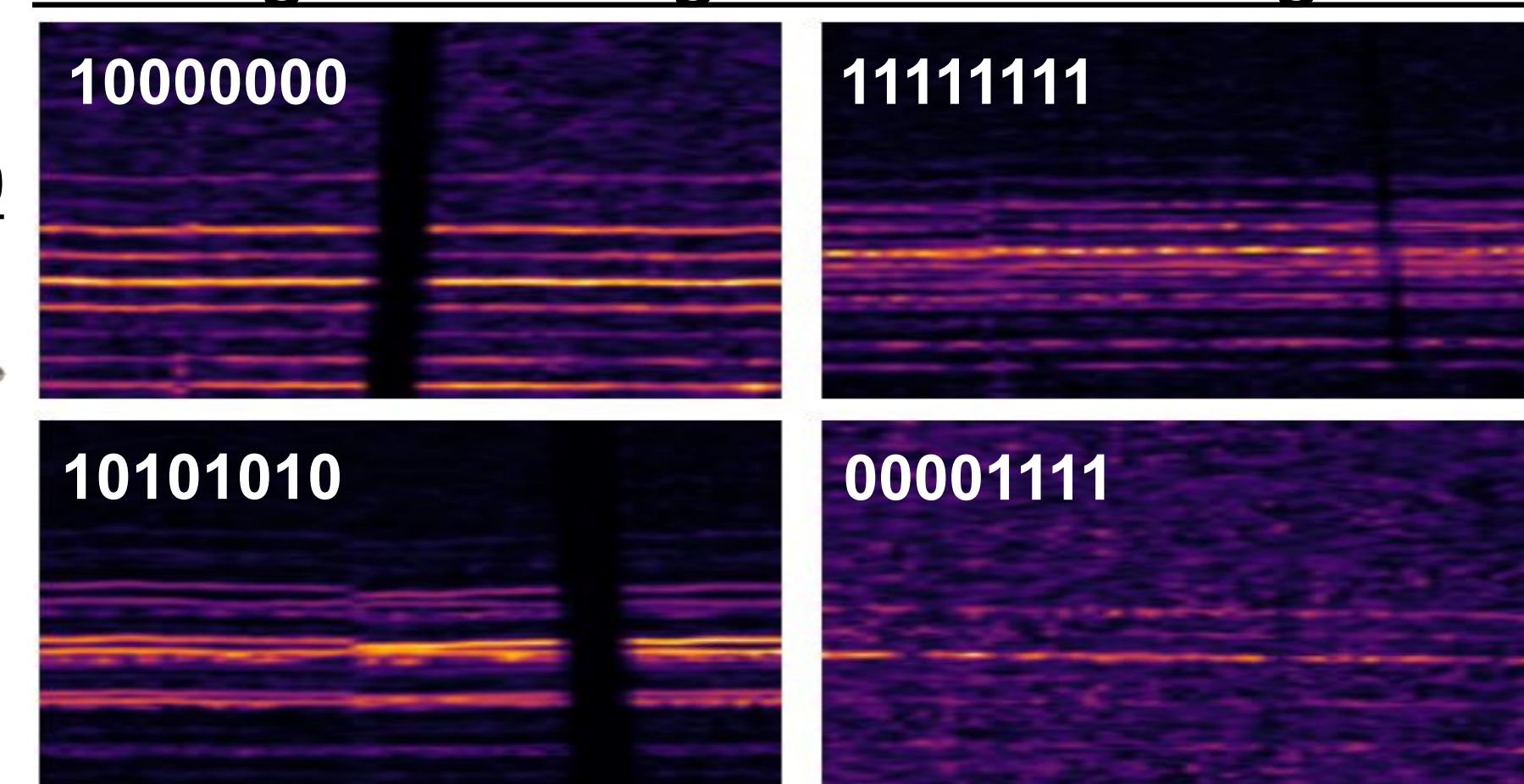
Augmentation & noise:

- Train:** random affine ($\pm 4^\circ/\pm 10\%$) + time-masking + Gaussian noise (0-32 dB SNR).
- Val:** mild affine ($\pm 1^\circ/\pm 2\%$), no mask, noise 0-32 dB.

Hold-out sets:

From the 10 % split, we created separate dataset files at fixed SNRs.

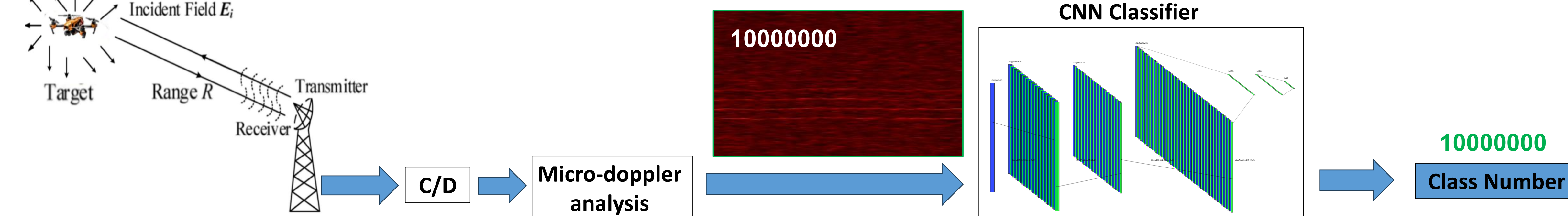
Training dataset augmentations of segements



CNN Model

We used three convolutional layers with 32, 64, and 128 filters, each followed by a ReLU activation and a 2x2 max-pooling layer. In the head, we applied dropout and a fully connected layer ending with a softmax activation.

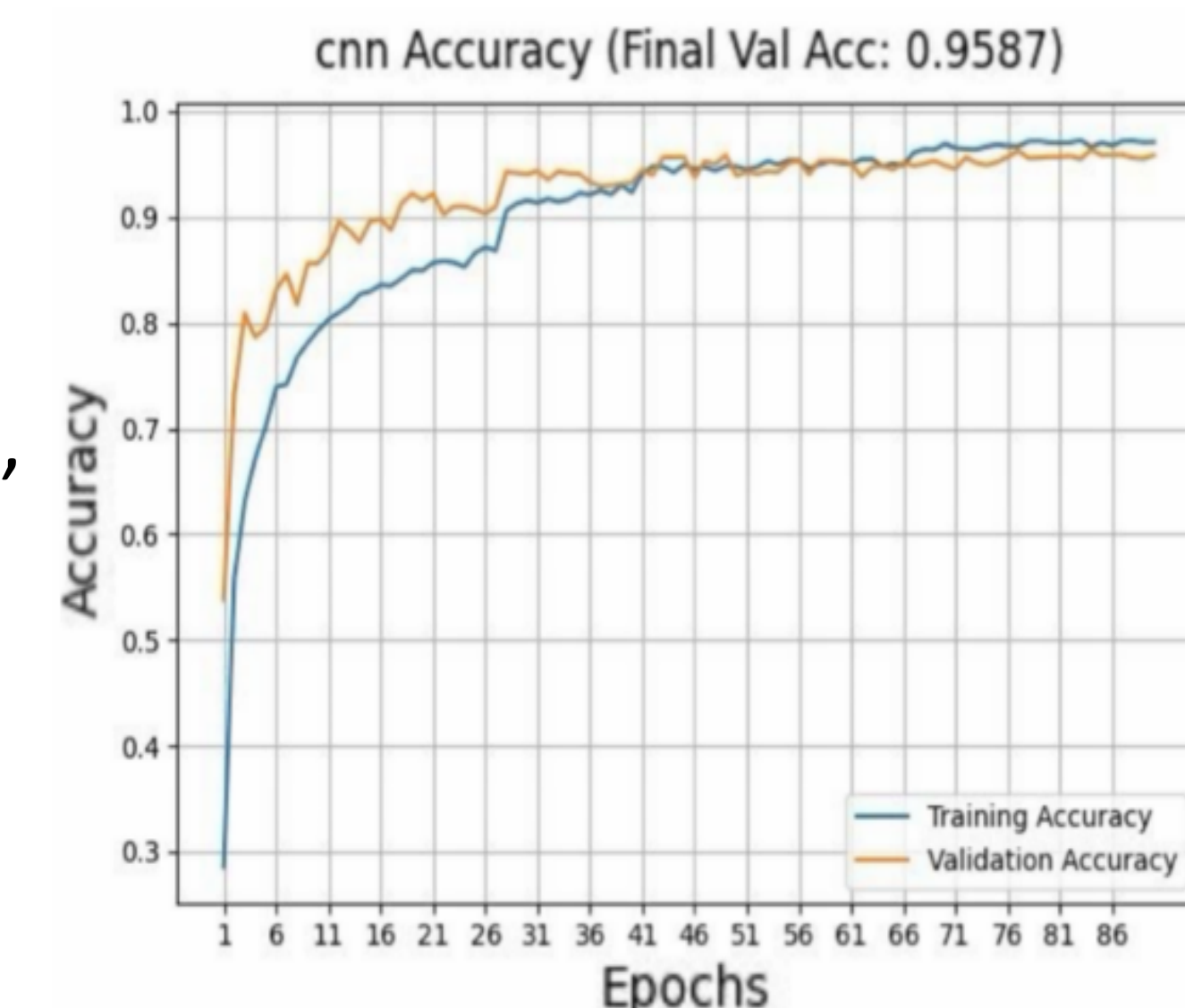
Project Block Scheme



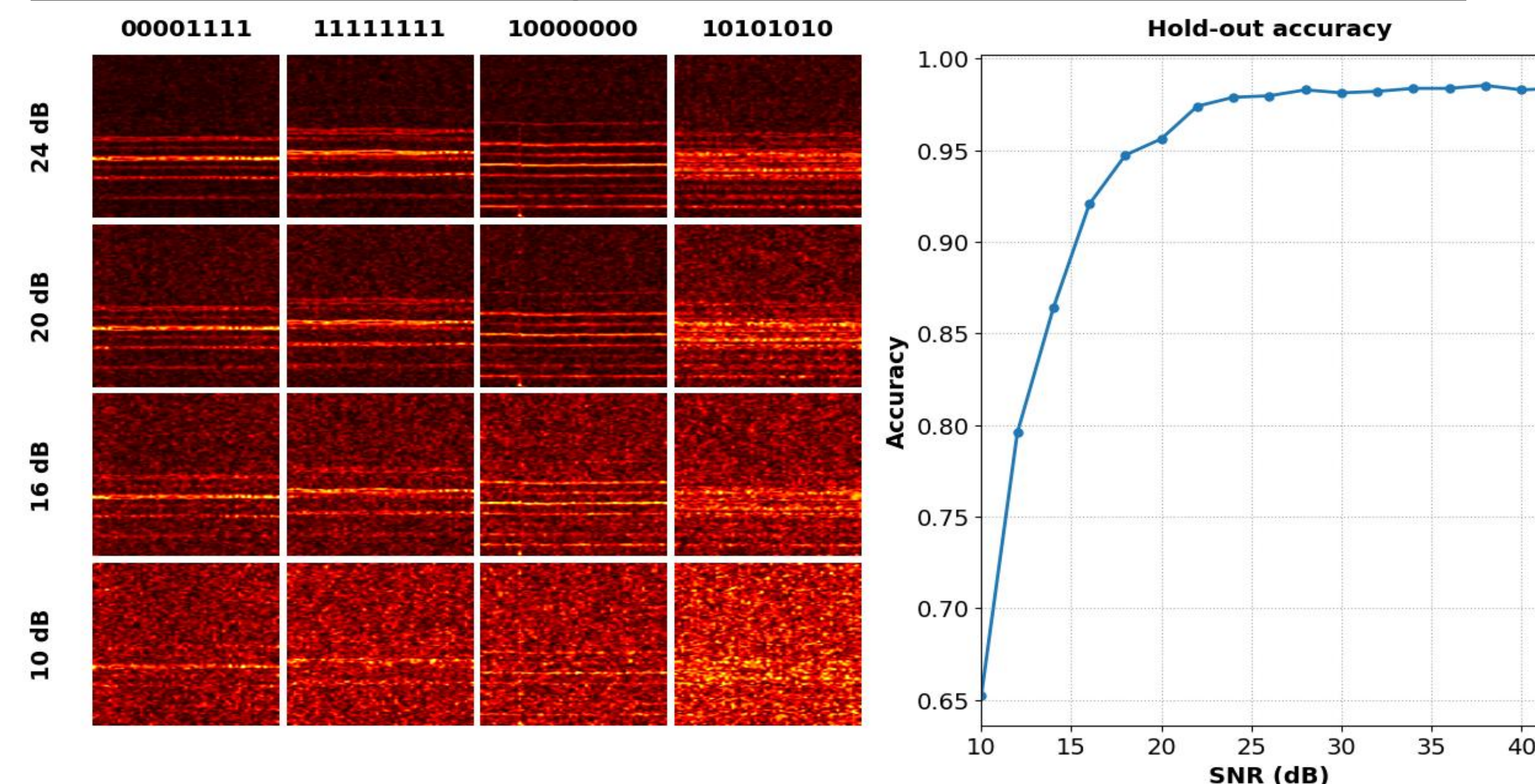
Results & Conclusions

Peak Validation Performance

After extensive hyperparameter tuning and repeated training runs, our CNN reached its highest validation accuracy at **90 epochs**, with both training and validation curves nearly overlapping. This close alignment indicates excellent generalization and minima overfitting.



CNN Validation Accuracy Across Datasets with Fixed SNR Levels



The accuracy curve demonstrates that our classifier exceeds 90 % once the SNR reaches 15 dB, and climbs to 97 % at 24 dB and above.

Bibliography

D. Vovchuk, M. Khobzei, V. Tkach, O. Eliashiv, O. Tzidki, K. Grotov, A. Glam, and P. Ginzburg, "Micro-Doppler-Coded Drone Identification," *arXiv:2402.04368*, Feb. 2024.